Practical Workbook

CT-261

DATABASE MANAGEMENT SYSTEMS

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Practical Workbook

Database Management Systems

(CT – 261)

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LAB#09

Constraints

Q1. Add a table-level PRIMARY KEY constraint to the EMP table on the ID column. The constraint should be named at creation. Name the constraint my_emp_id_pk.

Hint: The constraint is enabled as soon as the ALTER TABLE command executes successfully.

SOLUTION:

```
-- Q1

ALTER TABLE EMP
ADD CONSTRAINT my_emp_id_pk PRIMARY KEY(ID);
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table altered.
```

0.08 seconds

Q2. Create a PRIMARY KEY constraint to the DEPT table using the ID column. The constraint should be named at creation. Name the constraint my_deptid_pk .

Hint: The constraint is enabled as soon as the ALTER TABLE command executes successfully.

SOLUTION:

```
-- Q2

ALTER TABLE DEPT
ADD CONSTRAINT my_deptid_pk PRIMARY KEY (ID)
```

OUTPUT:

```
Results   Explain   Describe   Saved SQL   History

Table altered.

0.08 seconds
```

Q3. Add a column DEPT_ID to the EMP table. Add a foreign key reference on the EMP table that ensures that the employee is not assigned to a nonexistent department. Name the constraint my_emp_dept_id_fk.

SOLUTION:

Adding a column to DEPT_ID to EMP table:

```
-- Q3
ALTER TABLE EMP
ADD (DEPT_ID NUMBER)
```

OUTPUT:

```
Results   Explain   Describe   Saved SQL   History

Table altered.

0.08 seconds
```

Adding a Foreign Key Reference on EMP table:

```
ALTER TABLE EMP
ADD CONSTRAINT my_emp_dept_id_fk FOREIGN KEY(DEPT_ID)
REFERENCES DEPT(ID)
```

OUTPUT:

```
Results   Explain   Describe   Saved SQL   History

Table altered.

0.10 seconds
```

Q4. Confirm that the constraints were added by querying the USER_CONSTRAINTS view. Note the types and names of the constraints.

SOLUTION:

```
-- Q4

SELECT CONSTRAINT_TYPE,CONSTRAINT_NAME
FROM USER_CONSTRAINTS
WHERE TABLE_NAME IN ('EMP','DEPT');
```

OUTPUT:

Results Explain Describe Saved SQL History

CONSTRAINT_TYPE	CONSTRAINT_NAME
P	MY_EMP_ID_PK
R	MY_EMP_DEPT_ID_FK
P	MY_DEPTID_PK

Q5. Display the object names and types from the USER_OBJECTS data dictionary view for the EMP and DEPT tables. Notice that the new tables and a new index were created.

SOLUTION:

```
-- Q5

SELECT OBJECT_NAME,OBJECT_TYPE
FROM USER_OBJECTS
WHERE OBJECT_NAME IN('EMP','DEPT');
```

OUTPUT:

Results Explain Describe Saved SQL History

OBJECT_NAME	OBJECT_TYPE
DEPT	TABLE
EMP	TABLE

Q6. Modify the EMP table. Add a COMMISSION column of NUMBER data type, precision 2, scale 2. Add a constraint to the commission column that ensures that a commission value is greater than zero.

SOLUTION:

```
--Q6

ALTER TABLE EMP
ADD (COMMISSION NUMBER(2,2) CONSTRAINT emp_commission_check CHECK (COMMISSION>0));
```

OUTPUT:

Results | Explain | Describe | Saved SQL | History

```
Table altered.
```

0.08 seconds

```
DESCRIBE EMP;
```

Results | Explain | Describe | Saved SQL | History

Object Type **TABLE** Object **EMP**

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
EMP	ID	NUMBER	-	7	0	1	-	-	-
	LAST_NAME	VARCHAR2	25	-	-	-	✓	-	-
	SALARY	NUMBER	-	9	2	-	✓	-	-
	DEPT_ID	NUMBER	22	-	-	-	✓	-	-
	COMMISSION	NUMBER	-	2	2	-	✓	-	-